

ASSIGNMENT – 04

Q4. Biometric Attendance System (Face + Fingerprint) - Shruti Building

* Analyze the biometric attendance system as an IoT-enabled intelligent system. Clearly explain:

* The overall system architecture and working principle (data capture

-> processing

-> storage

-> access)

* The sensors and hardware components used for face recognition (camera, IR/depth sensors)

* The sensors and modules used for fingerprint (thumb impression) recognition (optical/capacitive fingerprint sensor)

* How both modalities are integrated (multi-modal biometrics) and how the system decides identity

* Data flow: acquisition -> feature extraction -> matching -> database/storage

* Networking, protocols, and backend/cloud interaction

* Key considerations: accuracy, latency, spoofing resistance, and security/privacy mechanisms

Introduction:

Arduino UNO R3 is a widely adopted microcontroller board employed in electronics studies and IoT applications. The Arduino UNO R3 consists of the ATmega328P microcontroller and serves as a straightforward microcontroller board to construct an embedded system.

In order to use the Arduino UNO R3 efficiently, it is crucial to comprehend the available documentation of the board like Pinout, Datasheet, Schematics, and it is the one that holds importance for beginners.

❖ Biometric Attendance System (Face + Fingerprint) – IoT Analysis

1. Overall System Architecture & Working Principle:

The system follows a typical IoT pipeline:

Data Flow Pipeline:

Data Capture → Processing → Storage → Access

❖ Explanation:

- Data Capture:

Face image (camera) + fingerprint scan (sensor)

- Processing:

Feature extraction using AI/ML algorithms

- Storage:

Biometric templates stored in local database or cloud

- Access:

Attendance marked and accessible via web/app dashboard and This creates a smart, automated, real-time attendance system

2. Face Recognition Hardware & Sensors:

❖ Components:

- ✓ Camera Module (RGB Camera)
- ✓ Captures facial image
- ✓ IR Sensor / Depth Sensor
- ✓ Detects 3D face structure
- ✓ Helps in low light conditions
- ✓ Prevents spoofing (photo/video attacks)
- ✓ Processor (Edge Device)
- ✓ Raspberry Pi / Embedded system / Microcontroller

- ✓ Runs face recognition algorithms

❖ **Working:**

Captures face → detects face → extracts features (eyes, nose, landmarks) → generates face embedding

3. Fingerprint Recognition Sensors & Modules:

❖ **Types of Sensors:**

- ✓ Optical Fingerprint Sensor
- ✓ Uses light to capture fingerprint image
- ✓ More affordable, widely used
- ✓ Capacitive Fingerprint Sensor
- ✓ Uses electrical signals
- ✓ More accurate and secure

❖ **Modules:**

- ✓ Fingerprint scanner module (e.g., R305, GT-511C3)
- ✓ Microcontroller interface (UART/USB)

❖ **Working:**

User places finger → sensor scans ridges & valleys → creates digital template

4. Multi-Modal Biometric Integration:

The system uses two biometric modalities:

- ✓ Face recognition
- ✓ Fingerprint recognition

❖ **Integration Methods:**

AND Logic (High Security):

Both face + fingerprint must match

OR Logic (Flexible):

Either one is sufficient

Weighted Fusion:

Assign confidence scores to both and combine

❖ **Decision Making:**

- ✓ Match scores are calculated
- ✓ If threshold is met → identity confirmed
- ✓ Attendance marked

❖ **multi-modal systems improve:**

- ✓ Accuracy
- ✓ Reliability
- ✓ Spoof resistance

5. Detailed Data Flow:

❖ **Step-by-Step:**

Acquisition:

Capture face image + fingerprint

Preprocessing:

- ✓ Noise removal
- ✓ Image enhancement

❖ **Feature Extraction:**

Face: embeddings (deep learning models like CNN)

Fingerprint: minutiae points (ridge endings, bifurcations)

Matching:

Compare extracted features with stored templates

Storage:

- ✓ Database (local or cloud)
- ✓ Stores templates, not raw images (for security)

Attendance Logging:

Timestamp + user ID recorded

6. Networking, Protocols & Backend:

❖ **Communication:**

Wi-Fi / Ethernet / Cellular

❖ **Protocols:**

HTTP/HTTPS – Web communication

MQTT – Lightweight IoT messaging

WebSocket – Real-time updates

❖ **Backend:**

Cloud server / Local server

Database (MySQL, Firebase, MongoDB)

❖ **User Interface:**

Web dashboard / Mobile app

Admin can view attendance reports

7. Key Considerations:

Accuracy

Multi-modal improves recognition rate

Reduces false positives/negatives

❖ **Latency**

Edge processing reduces delay

Fast response needed (<1–2 seconds)

❖ **Spoofing Resistance**

Face: IR/depth sensing, liveness detection

Fingerprint: capacitive sensors

❖ **Security & Privacy:**

- ✓ Encrypt data (HTTPS, SSL/TLS)
- ✓ Store templates instead of raw biometrics
- ✓ Access control & authentication
- ✓ Compliance with data protection policies

Conclusion:

This System is an IoT-enabled intelligent system because it:

- Uses sensors (camera + fingerprint) for data acquisition
- Processes data at edge devices
- Communicates with cloud/backend
- Provides real-time, automated decision-making
- Multi-modal biometrics ensures higher security, accuracy, and reliability, making it ideal for smart attendance in buildings like Shruti Building.